

Supplementary Material for Composite Silhouette

A. Extended Theoretical Analysis

Property 1

$$S_{\text{mM}}^{(b)} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b z_b}{2} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b}{2} \tanh\left(\frac{\Delta_b}{\Delta_{\text{max}} + \varepsilon}\right).$$

Proof. Starting from the definition of the subsample-specific composite score,

$$S_{\text{mM}}^{(b)} = w_b S_{\text{m}}^{(b)} + (1 - w_b) S_{\text{M}}^{(b)}.$$

Rewriting,

$$S_{\text{mM}}^{(b)} = w_b (S_{\text{m}}^{(b)} - S_{\text{M}}^{(b)}) + S_{\text{M}}^{(b)}.$$

Using $\Delta_b = S_{\text{m}}^{(b)} - S_{\text{M}}^{(b)}$ and $w_b = \frac{1+z_b}{2}$, we obtain

$$S_{\text{mM}}^{(b)} = \frac{1+z_b}{2} \Delta_b + S_{\text{M}}^{(b)} = \frac{\Delta_b}{2} + \frac{\Delta_b z_b}{2} + S_{\text{M}}^{(b)}.$$

Since $\Delta_b = S_{\text{m}}^{(b)} - S_{\text{M}}^{(b)}$,

$$\frac{\Delta_b}{2} + S_{\text{M}}^{(b)} = \frac{S_{\text{m}}^{(b)} - S_{\text{M}}^{(b)}}{2} + S_{\text{M}}^{(b)} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2}.$$

Therefore,

$$S_{\text{mM}}^{(b)} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b z_b}{2}.$$

Finally, substituting

$$z_b = \tanh\left(\frac{\Delta_b}{\Delta_{\text{max}} + \varepsilon}\right)$$

yields

$$S_{\text{mM}}^{(b)} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b}{2} \tanh\left(\frac{\Delta_b}{\Delta_{\text{max}} + \varepsilon}\right).$$

□

Property 2

$$S_{\text{mM}} = \frac{\bar{S}_{\text{m}} + \bar{S}_{\text{M}}}{2} + \frac{1}{2B} \sum_{b=1}^B \Delta_b z_b.$$

Proof. Starting from the definition of the Composite Silhouette score,

$$S_{\text{mM}} = \frac{1}{B} \sum_{b=1}^B \left[w_b S_{\text{m}}^{(b)} + (1 - w_b) S_{\text{M}}^{(b)} \right].$$

Using the subsample-level identity

$$w_b S_{\text{m}}^{(b)} + (1 - w_b) S_{\text{M}}^{(b)} = \frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b z_b}{2},$$

we obtain

$$S_{\text{mM}} = \frac{1}{B} \sum_{b=1}^B \left[\frac{S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)}}{2} + \frac{\Delta_b z_b}{2} \right].$$

Distributing the average gives

$$S_{\text{mM}} = \frac{1}{2B} \sum_{b=1}^B \left(S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)} \right) + \frac{1}{2B} \sum_{b=1}^B \Delta_b z_b.$$

By the definitions

$$\bar{S}_{\text{m}} = \frac{1}{B} \sum_{b=1}^B S_{\text{m}}^{(b)}, \quad \bar{S}_{\text{M}} = \frac{1}{B} \sum_{b=1}^B S_{\text{M}}^{(b)},$$

the first term becomes

$$\frac{1}{2B} \sum_{b=1}^B \left(S_{\text{m}}^{(b)} + S_{\text{M}}^{(b)} \right) = \frac{\bar{S}_{\text{m}} + \bar{S}_{\text{M}}}{2}.$$

Therefore,

$$S_{\text{mM}} = \frac{\bar{S}_{\text{m}} + \bar{S}_{\text{M}}}{2} + \frac{1}{2B} \sum_{b=1}^B \Delta_b z_b.$$

□

Probabilistic setup. For the probabilistic results below, we consider a fixed candidate number of clusters $k \in \mathcal{K}$. The subsamples are assumed to be drawn independently according to the sampling scheme, while any internal randomness of the clustering algorithm is also independent across trials and independent of the subsampling. Since, in our implemented method, the normalizing quantity $\Delta_{\text{max}}(k)$ is computed from the same subsamples used to form $S_{\text{mM}}(k)$, the resulting terms are not independent. To obtain concentration bounds with standard

tools, we therefore analyze a closely related version in which $\Delta_{\max}(k)$ is estimated from an independent set of subsamples and then used to form the final score. This preserves the form of the method while ensuring that the per-subsample composite scores used in the analysis are independent and identically distributed. In practice, this modification is mainly technical. The quantity $\Delta_{\max}(k)$ enters the method only through the normalization of the discrepancies, so it acts as a relative scale parameter rather than as a source of structural information on its own. Consequently, replacing the empirical $\Delta_{\max}(k)$ computed from the same subsamples by an independent estimate does not alter the form of the weighting rule, but only its normalization. When B is moderate to large, both quantities are expected to provide similar scaling of the discrepancies, and therefore to induce similar normalized values, weights, and final composite scores. The two versions are thus introduced to separate dependence for the sake of analysis, rather than because they represent different procedures.

A.1 Fixed- k concentration

Fix $k \in \mathcal{K}$, and define

$$\mu(k) = \mathbb{E} \left[S_{\text{mM}}^{(b)}(k) \mid \Delta_{\max}(k) \right].$$

Then, for any $t > 0$,

$$\mathbb{P}(|S_{\text{mM}}(k) - \mu(k)| \geq t \mid \Delta_{\max}(k)) \leq 2 \exp\left(-\frac{Bt^2}{2}\right).$$

Equivalently, for any $\delta \in (0, 1)$, with conditional probability at least $1 - \delta$,

$$|S_{\text{mM}}(k) - \mu(k)| \leq \sqrt{\frac{2 \ln(2/\delta)}{B}}.$$

Proof. Under our setup, the random variables

$$S_{\text{mM}}^{(1)}(k), \dots, S_{\text{mM}}^{(B)}(k)$$

are independent and identically distributed conditional on $\Delta_{\max}(k)$. Each of them lies in $[-1, 1]$. Hoeffding's inequality for bounded independent random variables therefore gives

$$\mathbb{P}\left(\left|\frac{1}{B} \sum_{b=1}^B S_{\text{mM}}^{(b)}(k) - \mu(k)\right| \geq t \mid \Delta_{\max}(k)\right) \leq 2 \exp\left(-\frac{Bt^2}{2}\right),$$

which is exactly the desired bound since

$$S_{\text{mM}}(k) = \frac{1}{B} \sum_{b=1}^B S_{\text{mM}}^{(b)}(k).$$

Solving

$$2 \exp\left(-\frac{Bt^2}{2}\right) = \delta$$

for t yields

$$t = \sqrt{\frac{2 \ln(2/\delta)}{B}},$$

which proves the claim. $\square$

A.2 Uniform concentration over the candidate set

Assume that the candidate set $\mathcal{K}$ is finite. Then, for any $\delta \in (0, 1)$,

$$\sup_{k \in \mathcal{K}} |S_{\text{mM}}(k) - \mu(k)| \leq \sqrt{\frac{2 \ln(2|\mathcal{K}|/\delta)}{B}}$$

with conditional probability at least $1 - \delta$.

Proof. Fix $t > 0$. By the fixed- k concentration result above, for each $k \in \mathcal{K}$,

$$\mathbb{P}(|S_{\text{mM}}(k) - \mu(k)| \geq t \mid \Delta_{\text{max}}(k)) \leq 2 \exp\left(-\frac{Bt^2}{2}\right).$$

Applying the union bound over $k \in \mathcal{K}$ gives

$$\mathbb{P}(\exists k \in \mathcal{K} : |S_{\text{mM}}(k) - \mu(k)| \geq t \mid \Delta_{\text{max}}(k)) \leq 2|\mathcal{K}| \exp\left(-\frac{Bt^2}{2}\right).$$

Setting the right-hand side equal to δ and solving for t yields

$$t = \sqrt{\frac{2 \ln(2|\mathcal{K}|/\delta)}{B}},$$

which proves the claim. $\square$

A.3 Recovery guarantee under a margin condition

Let

$$k^\dagger \in \arg \max_{k \in \mathcal{K}} \mu(k)$$

be an optimal candidate under the subsampling objective, and assume that this maximizer is unique with positive margin

$$\gamma := \mu(k^\dagger) - \max_{k \in \mathcal{K}, k \neq k^\dagger} \mu(k) > 0.$$

Let

$$k^* \in \arg \max_{k \in \mathcal{K}} S_{\text{mM}}(k)$$

denote the maximizer of the empirical Composite Silhouette score. If

$$B \geq \frac{8}{\gamma^2} \ln\left(\frac{2|\mathcal{K}|}{\delta}\right),$$

then

$$\mathbb{P}(k^\star = k^\dagger \mid \Delta_{\max}(k)) \geq 1 - \delta.$$

Proof. By A.2, with conditional probability at least $1 - \delta$,

$$\sup_{k \in \mathcal{K}} |S_{\text{mM}}(k) - \mu(k)| \leq \sqrt{\frac{2 \ln(2|\mathcal{K}|/\delta)}{B}}.$$

Suppose this event holds and let

$$\eta := \sqrt{\frac{2 \ln(2|\mathcal{K}|/\delta)}{B}}.$$

Then

$$S_{\text{mM}}(k^\dagger) \geq \mu(k^\dagger) - \eta,$$

while for every $k \neq k^\dagger$,

$$S_{\text{mM}}(k) \leq \mu(k) + \eta \leq \mu(k^\dagger) - \gamma + \eta.$$

Hence, if $\eta \leq \gamma/2$, then

$$S_{\text{mM}}(k^\dagger) \geq \mu(k^\dagger) - \eta \geq \mu(k^\dagger) - \gamma + \eta \geq S_{\text{mM}}(k)$$

for all $k \neq k^\dagger$, so $k^\dagger$ is the unique maximizer of $S_{\text{mM}}(k)$. The condition $\eta \leq \gamma/2$ is equivalent to

$$\sqrt{\frac{2 \ln(2|\mathcal{K}|/\delta)}{B}} \leq \frac{\gamma}{2},$$

which in turn gives

$$B \geq \frac{8}{\gamma^2} \ln\left(\frac{2|\mathcal{K}|}{\delta}\right).$$

Therefore, under this condition,

$$\mathbb{P}(k^\star = k^\dagger \mid \Delta_{\max}(k)) \geq 1 - \delta.$$

□

B. Transformations Ablation

We examine the sensitivity of Composite Silhouette to the choice of transformation applied to the normalized micro-macro discrepancy $\tilde{\Delta}_b$ before constructing the subsample-specific weight, while keeping the remaining pipeline fixed (including subsampling, k -means clustering, and Silhouette computation). In the

main method, we use $z_b = \tanh(\tilde{\Delta}_b)$, which yields the weight $w_b = (1 + z_b)/2$. To assess whether this choice is important in practice, we compare the proposed tanh transformation against three alternatives: a linear mapping, $z_b = \tilde{\Delta}_b$; a sigmoid transformation, implemented as $w_b = (1 + \exp(-\alpha \tilde{\Delta}_b))^{-1}$ with $\alpha = 1$; and a step-based transformation, which assigns full weight to the micro-averaged Silhouette when $\tilde{\Delta}_b > 0$ and full weight to the macro-averaged Silhouette otherwise. For each variant, we recompute S_{mM} and report the resulting selected number of clusters across the benchmark datasets. Table B.1 shows that the proposed tanh transformation is the most stable and accurate choice overall. In particular, it recovers the correct number of clusters on all datasets reported, whereas the alternative transformations yield several under- and over-estimations. This empirical pattern supports the role of tanh as a principled compromise: unlike the linear mapping, it smoothly compresses large discrepancies and thus avoids excessive sensitivity to extreme values; unlike the step rule, it preserves gradual adjustments in the relative weighting between micro- and macro-averaged Silhouette scores. We note that a sigmoid transformation with $\alpha = 2$ would be mathematically equivalent to the tanh weighting used in the main method, so in this ablation we use $\alpha = 1$ to obtain a genuinely distinct smooth alternative.

Table B.1: Estimated numbers of clusters via S_{mM} using various z_b transformations (original tanh, linear, sigmoid, step). **Bold values** indicate correct number of clusters selection, **red values** indicate suboptimal selection.

Dataset	S_{mM} k selection with z_b transformation			
	tanh	linear	sigmoid	step (sign)
S1	5	5	5	5
S2	6	3	6	3
S3	5	5	5	5
S4	12	13	12	13
PKS	2	2	3	2
WNE	3	3	4	3
HTR	2	2	2	2
DGT	10	8	10	10
BNK	2	2	2	2
NSG	20	20	20	21
SPM	2	2	2	2
STL	10	10	10	8
BBC	5	5	5	6
BLD	2	3	2	2
MDS	14	14	14	15
B77	77	72	77	77

C. Extended Empirical Validation

Table C.1: Values attained at the ground-truth number of clusters k_{GT} under the k -means protocol. Higher is better for S_{mM} , $\bar{S}_m$, $\bar{S}_M$, avg S_m , avg S_M , and avg CH, whereas lower is better for avg DB.

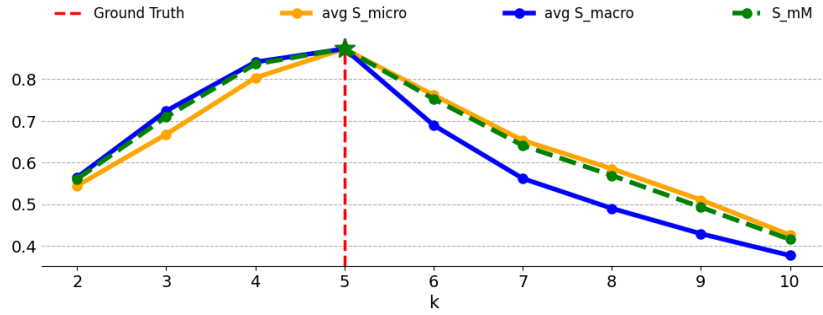
Dataset	S_{mM}	$\bar{S}_m$	$\bar{S}_M$	avg S_m	avg S_M	avg CH	avg DB
S1	0.8728	0.8727	0.8728	0.8728	0.8728	284172.5	0.1781
S2	0.6924	0.6924	0.6923	0.6924	0.6924	62284.8	0.4289
S3	0.6703	0.4767	0.7013	0.4851	0.7200	3879.3	0.4852
S4	0.6114	0.4236	0.6448	0.4211	0.6540	7214.3	0.5013
PKS	0.4047	0.4195	0.3221	0.4273	0.3287	99.7	1.0917
WNE	0.2875	0.2835	0.2887	0.2835	0.2886	70.6	1.3918
BLD	0.4186	0.4277	0.3681	0.4315	0.3689	429.4	1.0225
DGT	0.1848	0.1800	0.1861	0.1776	0.1819	165.0	1.9596
BBC	0.1111	0.1084	0.1114	0.1081	0.1109	160.9	2.6293
HTR	0.5874	0.6128	0.4444	0.6147	0.4509	9821.5	0.9174
STL	0.0292	0.0303	0.0244	0.0303	0.0225	136.3	3.6423
NSG	0.1996	0.0853	0.2222	0.0778	0.2019	321.0	1.8722
SPM	0.4320	0.4508	0.3658	0.3760	0.3084	245.1	2.0751
MDS	0.2689	0.2680	0.2681	0.2657	0.2654	45.6	1.7025
BNK	0.2325	0.2409	0.2091	0.2980	0.2314	6315.7	1.7660
B77	0.1720	0.1674	0.1736	0.1679	0.1758	281.4	1.9576

Table C.2: Selected number of clusters for S_{mM} , $\bar{S}_{\text{m}}$, $\bar{S}_{\text{M}}$, avg S_{m} , avg S_{M} , avg CH, and avg DB under a Bisecting k -means protocol. **Bold** entries indicate that the selected number of clusters is the ground-truth k_{GT} .

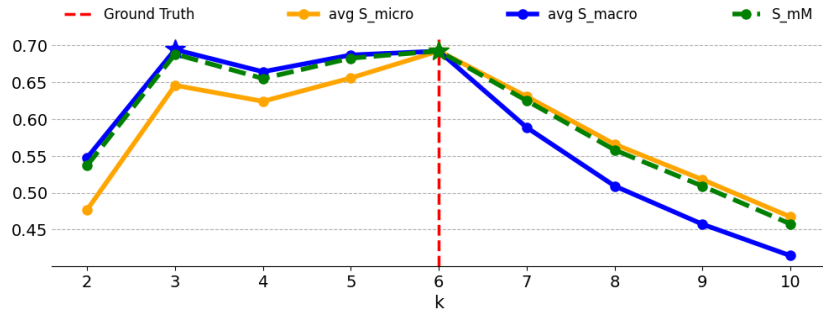
Dataset	S_{mM}	$\bar{S}_{\text{m}}$	$\bar{S}_{\text{M}}$	avg S_{m}	avg S_{M}	avg CH	avg DB
S1	5	5	5	5	5	5	5
S2	6	6	3	6	3	6	6
S3	3	2	5	3	3	10	5
S4	11	8	8	8	8	16	8
PKS	2	2	2	2	2	2	2
WNE	3	3	3	2	2	2	3
BLD	2	2	4	2	4	4	4
DGT	10	10	10	10	9	2	10
BBC	5	5	5	5	5	2	5
HTR	2	2	2	2	2	2	4
STL	10	10	9	12	8	6	8
NSG	19	15	19	15	23	15	24
SPM	2	2	2	2	2	3	7
MDS	15	15	14	16	13	13	14
BNK	2	2	2	2	2	3	8
B77	72	72	72	72	72	72	82

Table C.3: Selected number of clusters for S_{mM} , $\bar{S}_{\text{m}}$, $\bar{S}_{\text{M}}$, avg S_{m} , avg S_{M} , avg CH, and avg DB under a Gaussian mixture model protocol. **Bold** entries indicate that the selected number of clusters is the ground-truth k_{GT} .

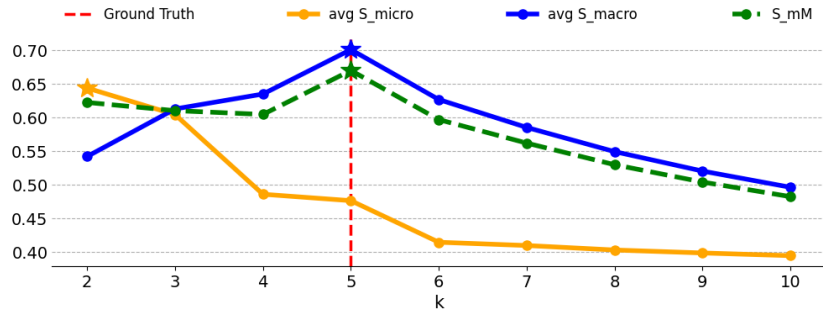
Dataset	S_{mM}	$\bar{S}_{\text{m}}$	$\bar{S}_{\text{M}}$	avg S_{m}	avg S_{M}	avg CH	avg DB
S1	5	5	5	5	5	5	5
S2	6	6	3	6	3	6	6
S3	3	3	3	3	5	10	5
S4	11	8	9	9	9	16	9
PKS	2	2	2	2	2	3	2
WNE	3	3	3	3	3	3	3
BLD	2	2	2	5	2	5	2
DGT	10	10	10	10	9	6	10
BBC	5	6	5	5	5	2	7
HTR	2	2	2	2	2	3	3
STL	10	10	10	6	6	6	10
NSG	24	24	24	24	24	24	24
SPM	2	2	2	2	2	2	2
MDS	14	15	12	15	14	14	14
BNK	2	2	2	2	2	2	2
B77	77	82	77	81	77	72	81



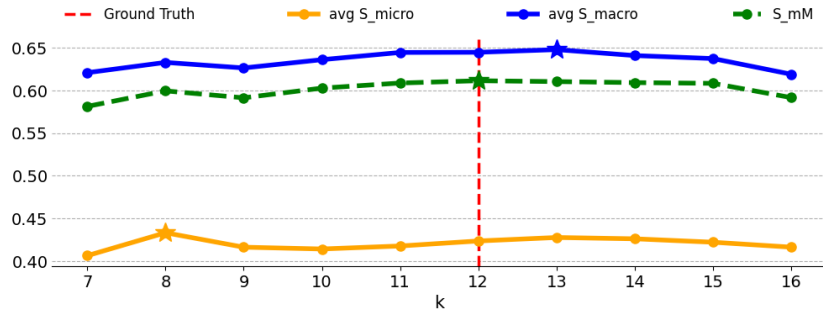
(a) S1



(b) S2

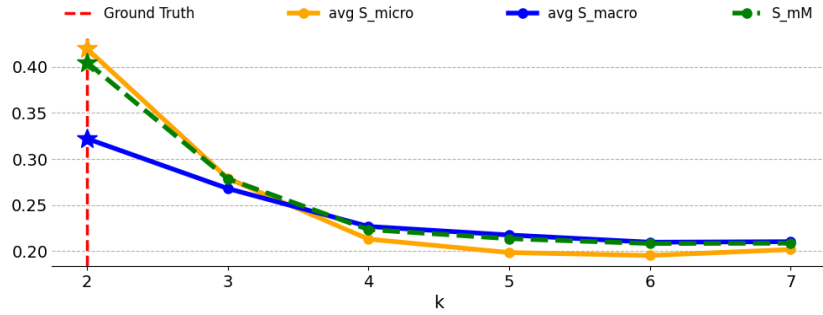


(c) S3

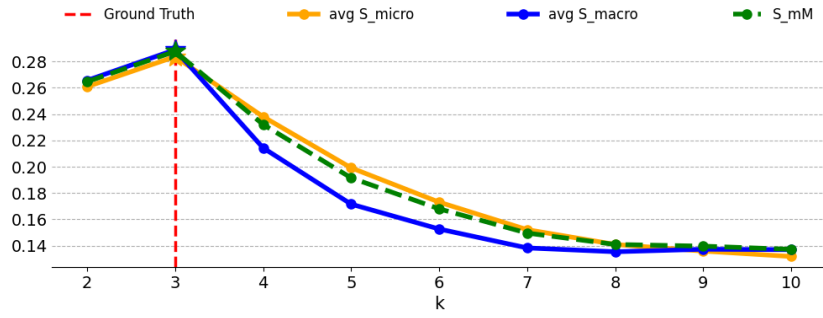


(d) S4

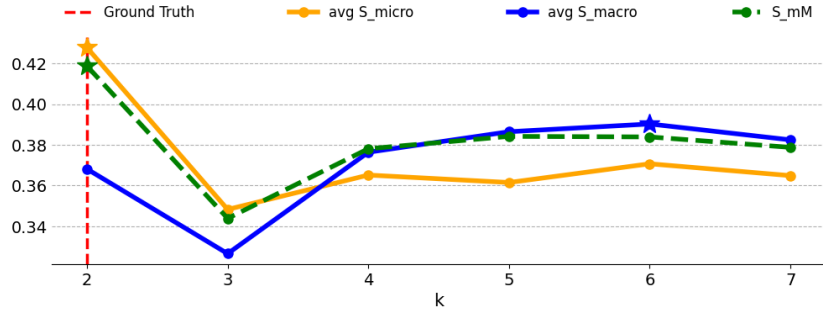
Fig. C.1: Trends of S_{mM} , $\overline{S}_m$, and $\overline{S}_M$ over the candidate number of clusters k for the four synthetic datasets. Stars ($\star$) denote the optimal cluster count selected by each criterion. The dashed line denotes the ground-truth number of clusters.



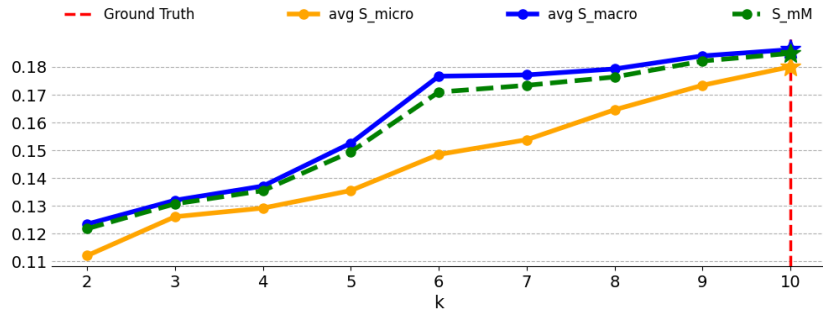
(a) PKs



(b) WNE

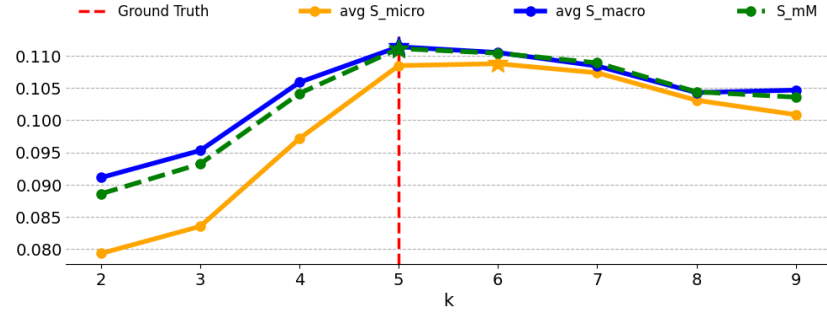


(c) BLD

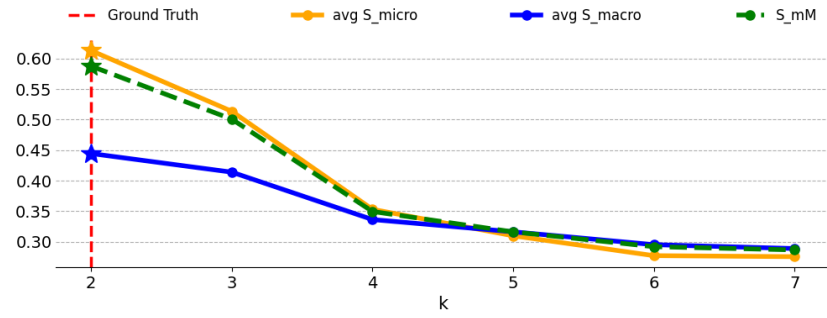


(d) DGT

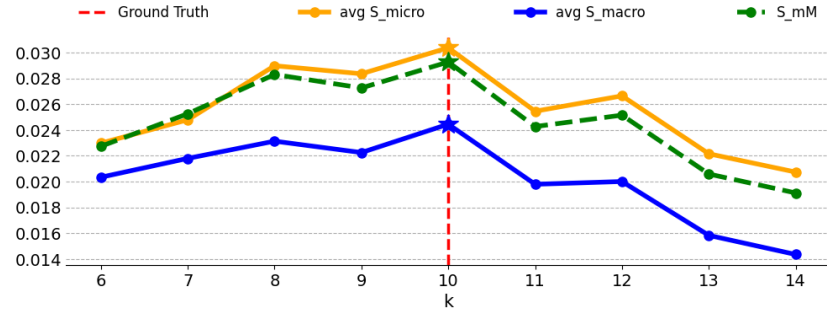
Fig. C.2: Trends of S_{mM} , $\overline{S}_m$, and $\overline{S}_M$ over the candidate number of clusters k . Stars (★) denote the optimal cluster count selected by each criterion. The dashed red line denotes the ground-truth number of clusters.



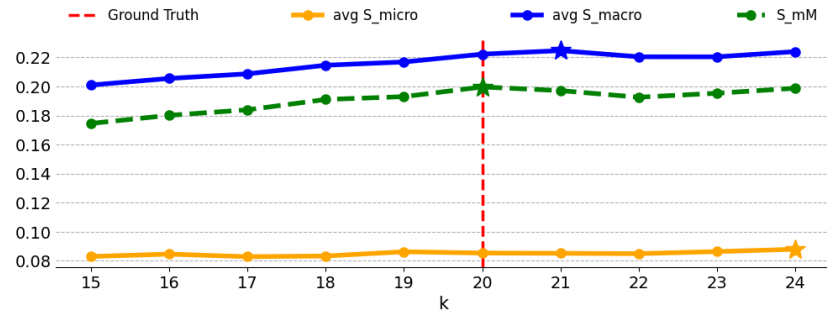
(a) BBC



(b) HTR



(c) STL



(d) NSG

Fig. C.3: Trends of S_{mM} , $\bar{S}_m$, and $\bar{S}_M$ over the candidate number of clusters k . Stars ($\star$) denote the optimal cluster count selected by each criterion. The dashed red line denotes the ground-truth number of clusters.

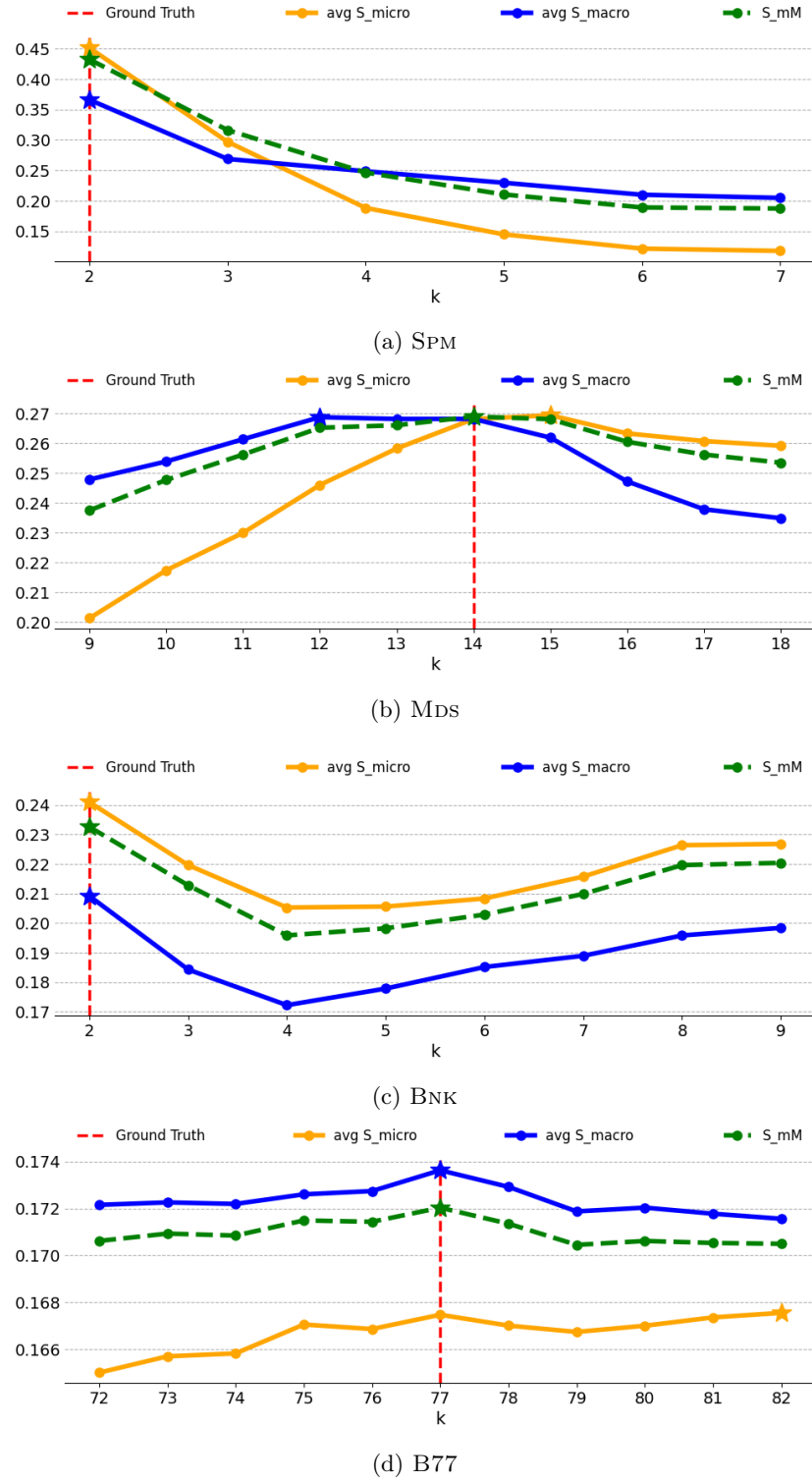


Fig. C.4: Trends of S_{mM} , $\bar{S}_m$, and $\bar{S}_M$ over the candidate number of clusters k . Stars ($\star$) denote the optimal cluster count selected by each criterion. The dashed red line denotes the ground-truth number of clusters.

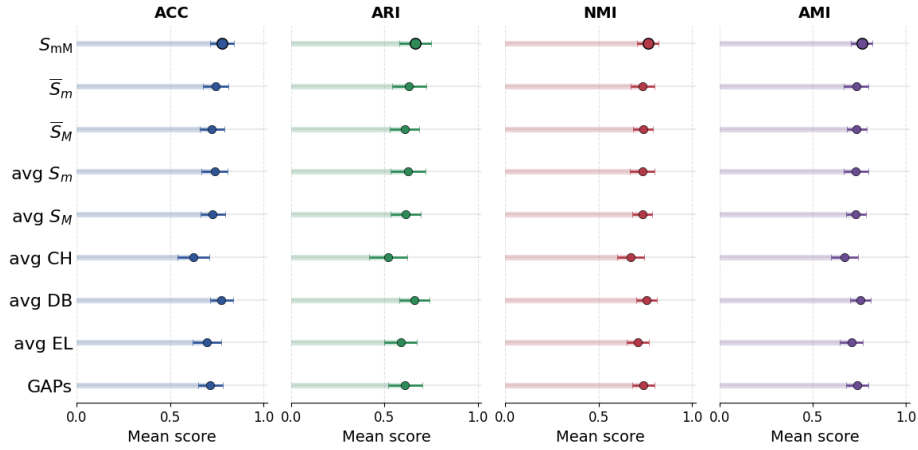


Fig. C.5: External clustering quality of the final full-data k -means partitions obtained at the number of clusters selected by each criterion. The analysis is performed on S1, S2, S3, S4, WNE, DGT, HTR, NSG, STL, BBC, MDS, and B77. For each dataset and criterion, we run full-data k -means using the criterion-selected value of k over multiple random initializations, and evaluate the resulting partitions against the labels. Dataset-level scores are first averaged over these repeated runs; points then denote the mean of these dataset-level averages across datasets, and horizontal intervals denote the standard error across datasets. ACC denotes clustering accuracy after optimal label matching; ARI, NMI, and AMI denote adjusted Rand index, normalized mutual information, and adjusted mutual information, respectively. The results show that selecting k with S_{mM} yields final partitions with consistently strong external quality, remaining competitive with or better than the strongest baselines across the reported metrics.